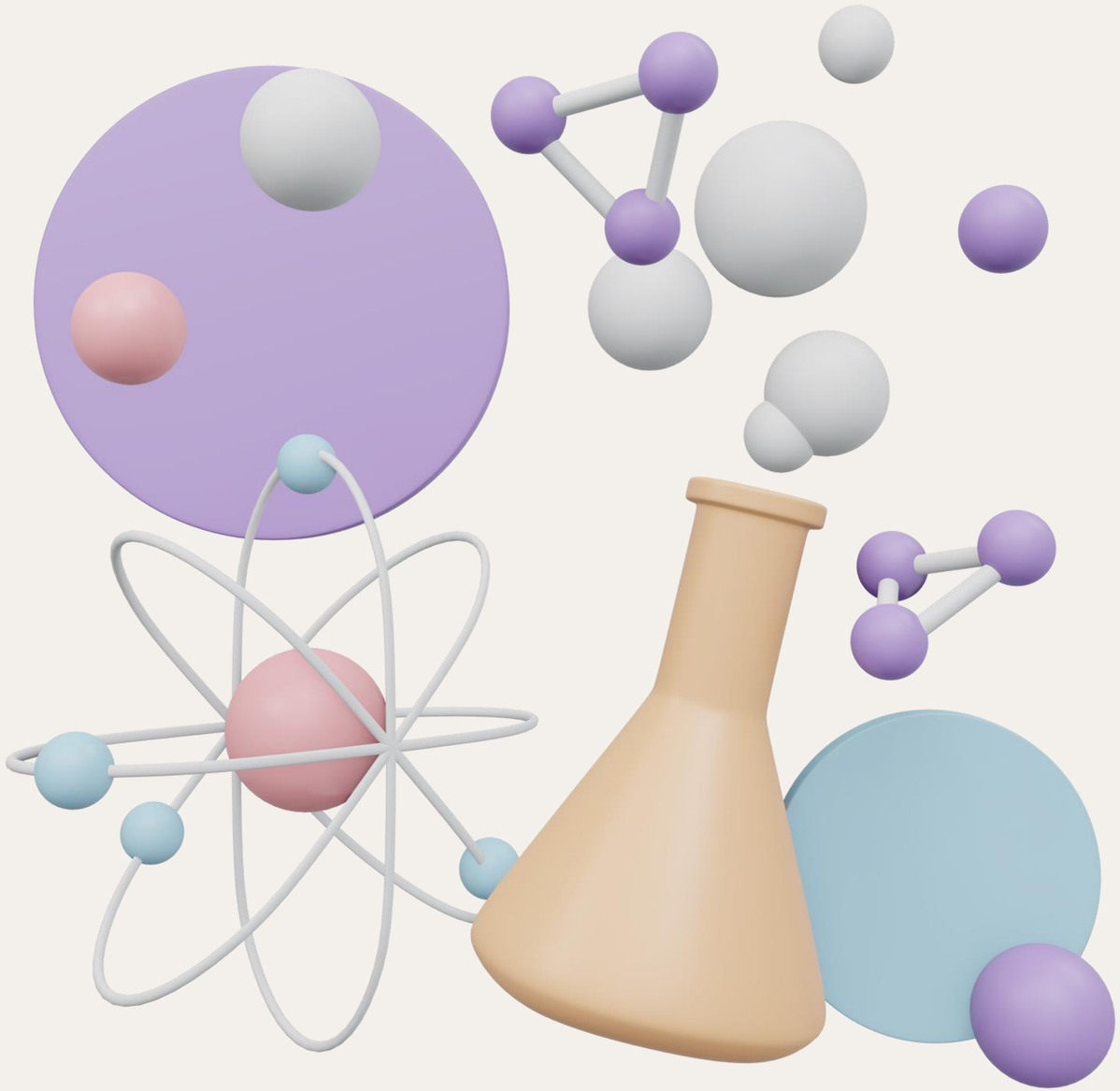


Expert Systems

MYCIN AND DENDRAL

ALCAZAR | AYAYO | BUERANO | DELA CRUZ | GRABATO | TABABAN

MYCIN

WHAT backward chaining

WHERE Standford University

WHY Assists in the diagnosis and suggestion of illnesses. In the field of organic chemistry, this approach aided in the discovery of novel chemical substances.

DENDRAL

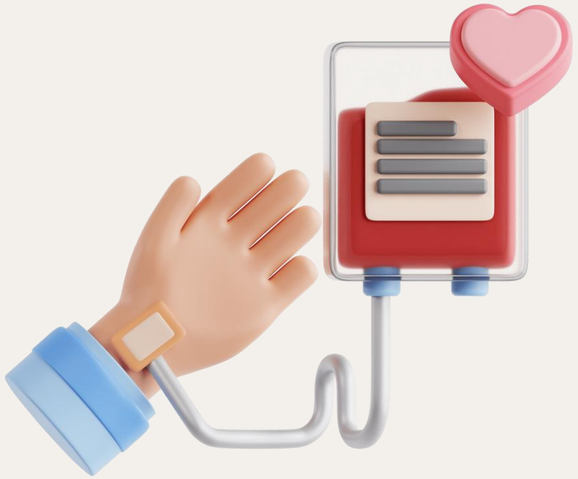
WHAT DENDRitic ALgorithm

WHERE Standford University, 1965

WHO Edward Feigenbaum and Joshua Lederberg

WHY Automates organic chemists' behavior in making decisions and solving problems, the DENDRAL program is regarded as the first expert system.

PROBLEMS THAT IT ADDRESSED



MYCIN

It is a computer-based consultation system designed to **assist physicians in the diagnosis of patients** with bacterial infections specifically in the blood.

DENDRAL

The primary aim of DENDRAL was to **help organic chemists in identifying unknown organic molecules** from compounds extracted from known origin that had medicinal or utility value.

THE EXPERTISE IT MIMICS

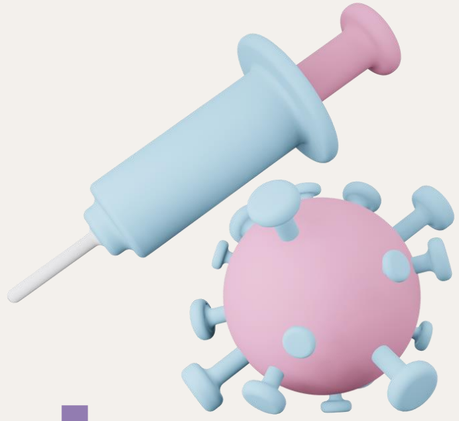
MYCIN

MYCIN emulates **physician** diagnosis expertise.

DENDRAL

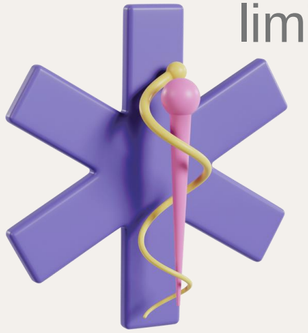
- DENDRAL mimics the expertise of a **chemist** by hypothesizing molecular structures from spectrographic data.

ADVANTAGES



MYCIN

- Detects severe bacterial infections like meningitis.
- Uses forward and backward chaining.
- Diagnosis examined in depth-first manner.
- System advantages: high accuracy, limited by human competency.



DENDRAL

- Uses a generate-and-test technique and a "plan-generate-test" policy.
- Performs comparably to human chemists and is utilized in industry and universities.
- The system's search approach has led to original research discoveries.

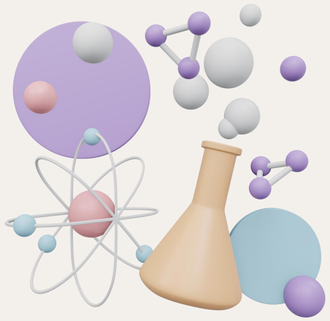
CHALLENGES

MYCIN

- Few medical professionals
- Ethical and legal issues
- Expensive
- Opaque

DENDRAL

- High Cost
- Difficulty of Interpretation
- Limited Scope



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